1. [Maximum mark: 6]

Evaluate the following

(*a*) $\int \left(\frac{2}{\sqrt[3]{x}} + \frac{5}{x^7} + x^{-\frac{5}{2}} \right) dx;$ [3]

(*b*) $\int \left(\frac{1}{\sqrt{x}} + 2\sqrt{x} \right)^2 \left(\sqrt[3]{x} + 1 \right) dx.$ [3]

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2. [Maximum mark: 6]

Evaluate the following

(a) $\displaystyle\int \frac{\sqrt{\ln(x)}}{x}\,dx$; [3]

(b) $\displaystyle\int 21e^{3x+2} - \cos(7x-11)\,dx$. [3]

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3. [Maximum mark: 5]

Let $a, b \in \mathbb{Z}^+$.

By evaluating the integral, show that

$$\int x\sqrt{ax^2 + b}\, dx = \frac{(ax^2 + b)^{3/2}}{3a} + C.$$

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4. [Maximum mark: 8]

Consider the function $f(x)$ such that

- $f''(x) = x^2 + 3$;
- The point $(1, 2)$ lies on $f(x)$;
- The tangent line to the graph of $y = f(x)$ at $x = 1$ is at an angle of $45°$.

Find an expression for $f(x)$.

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5. [Maximum mark: 10]

(a) Compute the following derivatives

(i) $\dfrac{d}{dx}\sqrt{x}$; [1]

(ii) $\dfrac{d}{dx}\sqrt{x}e^{\sqrt{x}}$. [3]

Let $I(x)$ denote the indefinite integral $\int e^{\sqrt{x}}\,dx$.

(b) Show your answer in part (a)(i) can be written as $\dfrac{e^{\sqrt{x}}}{2\sqrt{x}} + \dfrac{d}{dx}\dfrac{I(x)}{2}$. [2]

(c) Hence, solve for $I(x)$ by integrating both sides of

$$\frac{d}{dx}\sqrt{x}e^{\sqrt{x}} = \frac{e^{\sqrt{x}}}{2\sqrt{x}} + \frac{d}{dx}\frac{I(x)}{2}.$$ [4]

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